

Course B-Tech Semester Examination, 20 23 - 20 24

Class Test No. CA 3

Date 19/3/24

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Branch CSE

Paper Name Biology

Paper Code BSC 401



Signature of Invigilator

Group - A

1.(i) Carolus Linnaeus is considered as the father of Taxonomy

1.(ii) Two unicellular organisms are :-

a) Diatoms

b) Protista

1.(iii) R. H. Whittaker introduced the Five kingdom Classification

~~1.(iv)~~ -n in ~~1966~~ 1969.

1. (iv) False → Number of similar characters increases while we move from species to kingdom.

1.(v) Bentham and Hooker used natural system of classification

Group-B

2) The observations that led to major discoveries in the 18th and 19th century are as follows:-

(i) Microscopic Observation :-

Antonie Von Leeuwenhoek's pioneering work with microscopes ^{in 17th century} paved the way for 18th and 19th centuries. Scientists like Robert Hook observed and described cells for the 1st time leading to the cell theory.

(ii) Cell theory :-

cell theory proposed by Schwann and Virchow in the 19th century ~~stated~~ led to so many discoveries and observations.

The cell theory ~~is~~ states :-

- a) All living organism are made up of cells
- b) Cell is the basic structural and functional unit of all living organisms
- c) Cells ~~o~~ ~~p~~ exists from preexisting cell.

(iii) Theory of Evolution :-

In the 19th century, Charles Darwin's observation during his voyage on the Galapagos Island, led to the development of theory of evolution by natural selection.

3) Aspect	Prokaryotic	Eukaryotic
(i) Ribosome	The size of ribosomes in Prokaryotic cells are around 70S	The size of ribosomes in Eukaryotic cells are around 80S
(ii) Nucleus	Prokaryotic cell's nucleus don't have a membrane	Nucleus of Eukaryotic cell have two membrane

4) ~~Invent~~ Invention of camera is inspired by human eye. The statement can be justified by the following points.

(i) Aperture :- Human-eye can adjust the amount of ^{using} light entering the eye through the iris.

while the camera also contains mechanical structures called aperture that can change the amount of light.

(ii) lens :- In human-eye, a lens is present through which ~~is~~ the light is refracted. Similarly, the camera also contains lenses.

(iii) Screen :- In human-eye, the image is formed on the Retina. and in the camera also the image is formed on a screen at the back.

(iv) Processing :- In human-eye, the image is processed through the optical nerves while the camera ~~can~~ contains wires and mechanism for the processing.

5) *Mus musculus* and *Drosophila melanogaster* are considered as model because :-

- (i) Their rate of ~~pro~~ production is high so, research work is simple.

(ii) Their DNA structure is simple, which makes the study easy.

(iii) Their characteristics can be compared with humans. So, they ~~offe~~ are ~~to~~ useful for effective study.

iv) ~~They~~. They have high ^{enough} offspring to study the genetic mutation.

v)